Yile (Wayne) Wang

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Application: Student Researcher, PhD, Winter-Summer 2027

Dear Google Student Researcher Hiring Team,

I am a Ph.D. candidate in Cognition and Neuroscience at UT Dallas, with an expected graduation in 08/27. My research builds neural networks with biological learning rules to explain representational drift and the stability-plasticity problem in neural circuits, leading toward continual learning. I study which learning rules produce drift, aiming to turn biological learning rules into the next generation of machine learning (first-author paper under review at ICLR 2027). My dissertation asks how brain circuits extract invariants from drifting activity and weights, the question at the heart of continual learning.

I also created LLM-for-Zotero, an open-source AI research agent with 3,100+ GitHub stars and 650K+ downloads. I designed its adaptive agent loop, which plans, executes tools, and continues based on observed progress, and its retrieval over each user's own library, making it a personalized agent for academic research. For me, AI for science has two layers: building useful AI tools for scientists, and building brain-like AI systems that advance science itself. Neither should lag behind the other.

I would welcome projects on continual learning, neuroscience-inspired learning, or agents for science at Google Research or DeepMind, starting January to August 2027.

Sincerely,

Yile (Wayne) Wang